ARTIFICIAL INTELLIGENCE

Group Coding Assignment

Title: AI-Based Crop-Safe Path Planning for an Autonomous Potato-Field Robot

Activity	Group project	**Group size**	5–6 students
Language	Python	**Deadline**	**16 November 2026**

1. Background

An autonomous robot used for potato-field monitoring must travel between crop rows, visit selected monitoring locations and avoid damaging plants. It may also encounter obstacles such as stones, tools or other objects. In this assignment, you will model the field as a two-dimensional grid and use AI search techniques to plan a suitable route.

2. Problem Statement

Develop an intelligent path-planning agent that guides a robot from a specified starting position to a destination in a simulated potato field. The robot must avoid obstacles and crop-protection regions while selecting a route with a low overall movement cost. Your program should also demonstrate how the robot can update its knowledge of the field when it detects an obstacle that was not known in advance.

3. Tasks to be Completed

- **Task 1 – Field environment:** Build a 2D grid for a potato field. Mark safe travelling cells, potato rows, crop-protection regions, obstacles, the robot's start position and its destination. Use a clear visualisation.
- **Task 2 – Problem formulation:** Specify the initial state, available actions, transition model, goal test and path-cost function. State how the robot is allowed to move.
- **Task 3 – Search algorithms:** Implement Breadth-First Search (BFS), Depth-First Search (DFS), Greedy Best-First Search and A* Search. Define and explain the heuristic used by the informed algorithms.
- **Task 4 – Crop-safe planning:** Ensure that the robot does not enter obstacle cells or crop-protection cells. Introduce a suitable cost or penalty for routes that pass too close to protected crop regions, so safer alternatives are preferred where available.
- **Task 5 – Obstacle detection and replanning:** Start with a map containing at least one obstacle unknown to the robot. During the simulated journey, let the robot detect it, update its known map and use A* to calculate a new route if the current route is blocked. Detection can be simulated in Python; hardware or actual sensors are not required.
- **Task 6 – Performance evaluation:** Compare algorithms using path length, number of expanded nodes, execution time and whether the destination was reached. For the online-obstacle experiment, also report the number of replanning operations. Present the results in a table.

4. Expected Program Output

- A visual display of the field grid, crop rows, protected regions and obstacles.
- The robot's starting position and destination.
- The path generated by each implemented search algorithm.
- A demonstration of an initially unknown obstacle being detected and the route being replanned.
- A comparison table summarising the performance of the algorithms.

5. Group Work

Each group must have 5–6 students. The following division of work is suggested. Groups may adjust it according to their team size.

Member	Suggested responsibility
1	Field environment design and problem formulation
2	BFS and DFS implementation
3	Greedy Best-First Search and heuristic design
4	A* implementation and crop-safe path cost
5	Obstacle detection simulation, map update and replanning
6 (if applicable)	Testing, visualisation, performance comparison and documentation

All members are expected to contribute and should be able to explain the overall approach and the main parts of the code.

6. GitHub Repository and Submission

Each group must create a GitHub repository and upload its project work. The repository must be accessible to the instructor.

The repository should include:

- Complete Python source code.
- A README.md file explaining the project, algorithms, required libraries and instructions to run the program.
- Screenshots of the program output.
- The performance comparison table and a brief summary of observations.
- Names and roll numbers of all group members, with each member's contribution.

Suggested repository name: AI_CropSafe_PathPlanning_GroupXX

Email submission: Send one email per group with the following details.

To	jchakole@iiitn.ac.in
Subject	AI Group Coding Assignment Submission – Group XX
Include	Group number; names and roll numbers of all members; GitHub repository link; and name/contact details of the group representative.

Last date for submission: 16 November 2026. Please check that the repository link works and that the instructor can access the uploaded files before the deadline.

7. General Instructions

- Use Python. Libraries such as NumPy and Matplotlib may be used.
- The purpose is to apply the AI search techniques discussed in class. You must understand and be able to explain your implementation.
- Do not submit code copied from another group or an online source without understanding it. Acknowledge any external references or code used.
- Every group member should be prepared to explain the project during evaluation.

The focus of this assignment is to apply the AI concepts discussed in class to a practical path-planning problem.